Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	EV Market Performance and Insights Dashboard (Tableau)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset with no live data refresh.	Insights become outdated as new sales and battery data arrive.	High
2	Many vehicle categories (14+) make some charts crowded.	Harder for executives to read quickly on small screens.	Medium
3	No forecasting or predictive analysis.	Cannot project future sales, range or adoption trends.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Tableau workbook.	Publish to Tableau Server / Cloud for many concurrent viewers with role-based access.
2	Data Storage	Static file-based dataset.	Move to a cloud database (SQL / data warehouse) with scheduled extract refresh.
3	Performance	Smooth on current data size.	Use extracts, aggregated data and context filters to handle millions of rows.
4	Security	Local file, no access control.	Add user authentication, row-level security and permission groups.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Connect live data source with automatic refresh and add more years of data.	1-2 months	Always up-to-date insights for decision makers.
Phase 3	Add forecasting models and what-if parameters for sales and range.	3-4 months	Supports future product and market planning.
Phase 4	Publish web / mobile dashboards with alerts; add charging infrastructure and policy data.	6 months	Wider access for investors, researchers and policymakers.